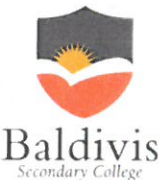


Name:	Answers		Date:	
Teacher:	*POWER*			
	Year 9 Mathematics Mini test 4 Topic – Data analysis and Displays		SCORE: / 37	
	<u>Full working out MUST be shown to get full marks for each question.</u>			
Total Time:	40 minutes			
Weighting:	5 %			
Equipment:	To be provided by the student: Pens and Pencils, scientific calculator, A4 page of notes.			

- 1) Callum measures the minutes it takes it him to get to school in the morning. He records the data in the table below [7 Marks]

M	T	W	T	F	M	T	W	T	F
15	24	27	12	18	18	15	23	24	19

- a. From the data above, on which days do you think Callum walked to school? [1]

Tuesday, Wednesday, Wednesday 2, Thursday ✓ all 4 days

- b. What was Callum's average time to get to school? [2]

$$\frac{\Sigma}{n} = \frac{195}{10} = 19.5 \text{ minutes Ans}$$

Divided Sum by 10

- c. What was the Median time it took Callum to get to school? [2]

12, 15, 15, 18, 18, 19, 23, 24, 24, 27 ✓ order

18.5 minutes ✓ median

- d. Comparing the average time, to the median time, which do you believe is a more accurate representation of his usual travel time to school? why? [2]

Both are similar, but because the times 18.5 is closer to 19, 60% of the time he is sub 20 minutes. ✓✓ Reasonable Response

- 2) Georgia received the following scores for her last 4 tests – 56%, 69%, 82%, 89%. She wants to maintain an average of 75% percent. what score would she need on her final test to achieve this average? [3 Marks]

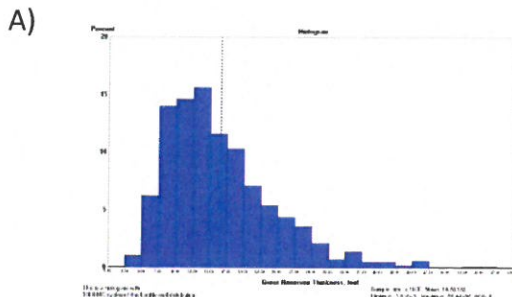
$$75 \times 5 = 375 - (56 + 69 + 82 + 89)$$

$$375 - 296 \text{ Subtract}$$

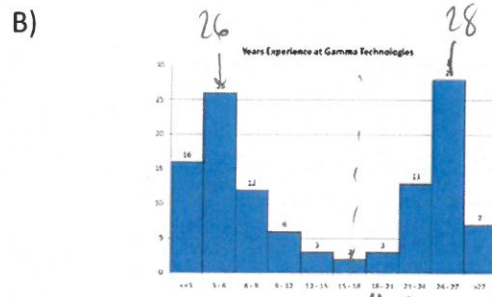
$$= 79 \% \text{ Ans}$$

3) The shapes of the graphs below can be described using the words "Positive Skew", "Negative Skew", "Symmetrical" and "Bimodal". Underneath each graph, write the most appropriate term.

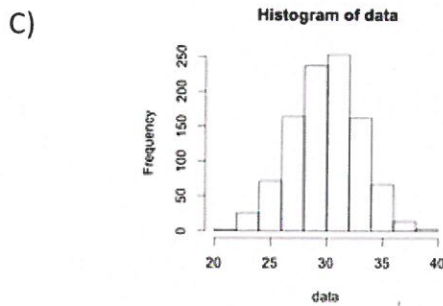
[6 marks]



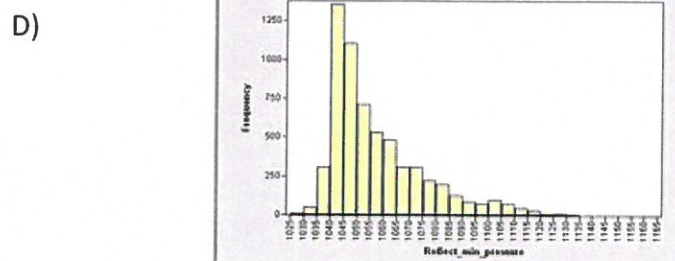
Positive Skew ✓



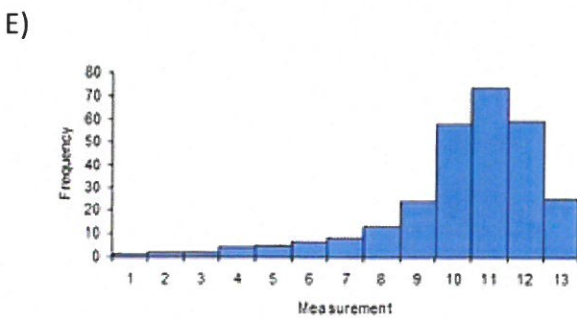
Symmetrical ✓



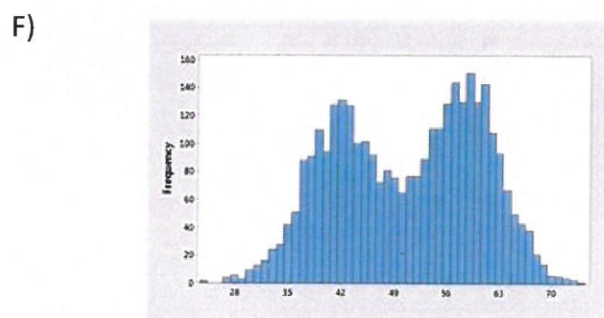
Symmetrical ✓



Positive Skew ✓



Negative Skew ✓



Symmetrical ✓

- 4) Sam and Anna read in book "there are 3 times more white cars than black". They test this claim by surveying their school car parks and see that there is only twice as many white cars than black cars. They come to the conclusion that the books information is wrong. [5 marks]

a. Was the books information first or second hand?

[1]

Second hand ✓

b. List two reasons as to why Sam and Anna may have gotten different results to the book?

[2]

→ Number of People at school ✓
→ only 2 to 1 white to black } Anything Reason

c. With relation to population size, do you think that Sam and Anna may have conducted a fair survey to base their belief's on, or should they reconsider that the book may be correct?

Dependly on how many cars they Surveyed, [2]
they may have had 4 cars, with 2 white,
one black and a Silver, or 30 with
different denominations.

By Not knowing the population of the Sample,
we Cannot be Sure at No Bias

✓ mention bias or not w/ reason
✓ me

✓ mentions bias

✓ provides reasoning

5) The data below is information about the number of hours worked by different employees during 1 week at Built Fast Enterprises. [5 + 4 + 5 + 2 = 16 marks]

Week 1 data. 6, 7, 9, 13, 17, 19, 19, 24, 29, 29, 33, 35, 37, 38, 40, 42, 46, 60, 62

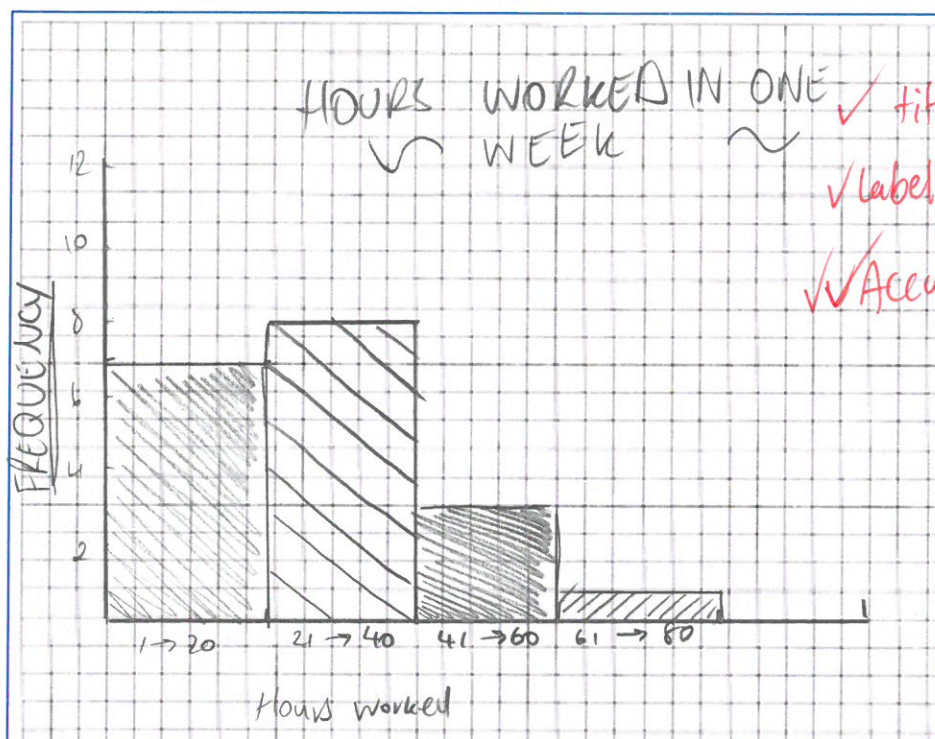
a. create a frequency table for this data.

Interval	Tally	Frequency
1 → 20		7
21 → 40		8
41 → 60		3
61 → 80		1
Total		19

Layout + title
Intervals
tallys
total

5

b. Use the frequency table to create a histogram.



✓ title
✓ label on Axes (- 1/2 for each missing)
✓ Accurate

4

c. The company decided to compare the week 1 data from above, to the week after. Create a back to back stem and leaf plot using the data provided.

Week 2 data. 9, 15, 16, 16, 17, 19, 23, 24, 29, 29, 29, 34, 35, 35, 39, 41, 41, 51, 55

leaf wk1	stem	leaf wk2
9 7 6	0	9
9 9 7 3	1	5 6 6 7 9
9 9 4	2	3 4 9 9 9
8 7 5 3	3	4 5 5 9
6 2 0	4	1 1
	5	1 5
2 0	6	

✓ back to back
✓ ordered
✓ Key
✓ labelled sides

3

d. Which week were the employees the most productive? Use evidence from your stem and leaf plot in your explanation.

Week 1 people stayed longer, and the data is more positively skewed and symmetrical

✓ week 1
✓ valid reason

2